

MULTI-TYPE BIRTH-DEATH MODELS

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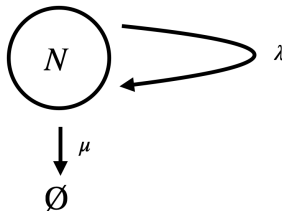
Day 4

- 1 MOVING TO A BAYESIAN FRAMEWORK
- 2 MULTI-TYPE BIRTH DEATH MODELS
- 3 SIMULATING MODELS IN R AND BEAST2

TABLE OF CONTENTS

- 1 MOVING TO A BAYESIAN FRAMEWORK
- 2 MULTI-TYPE BIRTH DEATH MODELS
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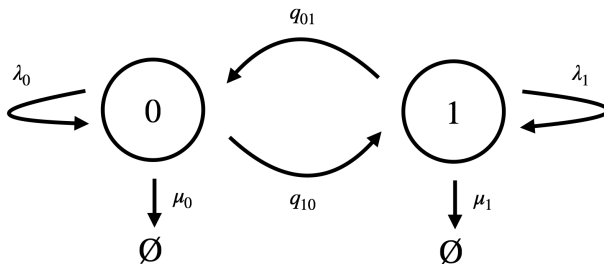
RECALL: BIRTH-DEATH MODELS AND BISSE MODELS



In the basic birth-death model, individual lineages can replicate or die

The parameter of this model is just $\theta = (\lambda, \mu)^T$, the replication/diversification rate, and the death/extinction rate

RECALL: BIRTH-DEATH MODELS AND BiSSE MODELS



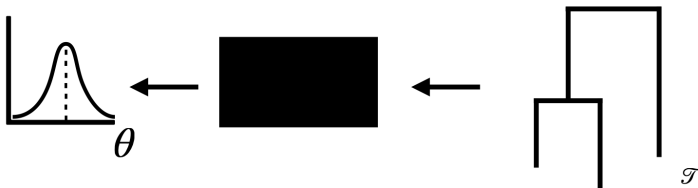
In the BiSSE model, trees $\mathcal{T}$ depend on the parameters $\theta = (\lambda_0, \lambda_1, \mu_0, \mu_1, q_{01}, q_{10})^T$

WHAT FACTORS SHAPE TREES?



In the forward direction, parameters, θ , of these models dictate the types of trees we are likely to see

HOW MUCH INFORMATION ABOUT PARAMETERS IS CONTAINED IN TREES?



In the other direction, the trees we observe contain information about the parameters that created them (assuming the tree-generating model is appropriately specified)

PHYLODYNAMIC LIKELIHOOD

$$p(\theta, \mathbf{Q}, \mathbf{A} | \mathbf{s}, \mathbf{x}) \\ \propto \int_{\mathcal{T}} \sum_{\mathbf{u}, \mathbf{y}} p(\mathbf{s}, \mathbf{x}, \mathbf{u}, \mathbf{y} | \mathcal{T}) p(\mathcal{T} | \theta) p(\theta) p(\mathbf{Q}) p(\mathbf{A})$$

The general approach in BEAST2 is to adopt a Bayesian approach by supplying priors on the parameters of the tree-generating model, the sequence evolution model, and the character evolution model, and perform inference on the posterior distribution

PHYLODYNAMIC LIKELIHOOD

In BEAST2 parlance, models for tree generation such as the Yule model, birth-death model, and their state-dependent variants are called “tree priors”

BiSSE models are accommodated in BEAST2 as special cases of more general “multi-type birth death models”, which we dive into in the next section

Another class of models available are the **structured coalescent** models, which we will not discuss

Multi-type birth death and structured coalescent models are the two main classes of phylodynamics models that are included in BEAST2

TABLE OF CONTENTS

- 1 MOVING TO A BAYESIAN FRAMEWORK
- 2 MULTI-TYPE BIRTH DEATH MODELS
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BASIC MULTI-TYPE BIRTH DEATH MODELS

We will introduce the most basic multi-type birth-death model in these slides.

You should be able to appreciate that generalizing these models is fairly straightforward to do (although calculating the likelihoods can become increasingly cumbersome)

Basic multi-type birth death model: each type can give birth to other types

- can ignore migration, or include it
- can include a sampling rate to produce heterochronous trees
- result: a character-mapped tree (just like SSE models)

MULTI-TYPE BIRTH DEATH MODELS¹

λ_{ij} : rate at which individuals in i produce individuals in j

d_i : death rate of individuals in state i

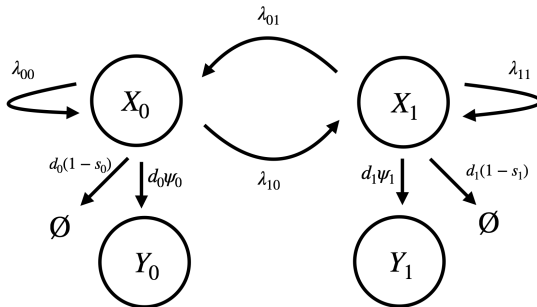
s_i : probability of sampling upon death

This will produce a tree with sampled and unsampled individuals - only the sampled individuals are included in the final tree

So, this is very similar in spirit to the SSE models we have seen before, but it does not have separate migration terms q_{ij} - state changes always coincide with branching events in trees. You immediately see how to modify the model to incorporate these effects.

¹Tanja Stadler and Sebastian Bonhoeffer. "Uncovering epidemiological dynamics in heterogeneous host populations using phylogenetic methods". In: *Philosophical Transactions of the Royal Society B: Biological Sciences* 368.1614 (2013), p. 20120198.

MULTI-TYPE BIRTH DEATH MODELS²



X_i unobserved, Y_i observed

²Tanja Stadler and Sebastian Bonhoeffer. "Uncovering epidemiological dynamics in heterogeneous host populations using phylogenetic methods". In: *Philosophical Transactions of the Royal Society B: Biological Sciences* 368.1614 (2013), p. 20120198.

MULTI-TYPE BIRTH DEATH MODELS³

Much like BiSSE, we derive a backward equation for $D_{Ni}(t)$, the probability that we observe the tree below branch N (node N is at the tipward end of branch N), given that branch N is in state i at time t :

$$dN_i(t + \Delta t) = \left(1 - \left(\sum_j \lambda_{ij} + d_i \right) \Delta t \right) D_{Ni}(t) + 2 \sum_j \lambda_{ij} \Delta t E_j(t) D_{Ni}(t)$$

³Tanja Stadler and Sebastian Bonhoeffer. “Uncovering epidemiological dynamics in heterogeneous host populations using phylogenetic methods”. In: *Philosophical Transactions of the Royal Society B: Biological Sciences* 368.1614 (2013), p. 20120198.

MULTI-TYPE BIRTH DEATH MODELS⁴

Likewise, we have to track extinction probabilities,

$$\begin{aligned} dE_i(t + \Delta t) = & (1 - s_i) d_i \Delta t \\ & + \left(1 - \left(\sum_j \lambda_{ij} + d_i \right) \Delta t \right) E_i(t) \\ & + \sum_j \lambda_{ij} \Delta t E_i(t) E_j(t) \end{aligned}$$

Notice how $(1 - s_i)$ and d_i are multiplied in this equation

⁴Tanja Stadler and Sebastian Bonhoeffer. “Uncovering epidemiological dynamics in heterogeneous host populations using phylogenetic methods”. In: *Philosophical Transactions of the Royal Society B: Biological Sciences* 368.1614 (2013), p. 20120198.

MULTI-TYPE BIRTH DEATH MODELS⁵

Observations at tips supply initial conditions for our backward equations: the present is time 0, and because tips are sampled through time, we observe a tip at time τ in the past (time measured going into the past)

$$D_{Ni}(\tau) = d_i s_i$$

if the tip is in state i , and equals zero otherwise

Notice that s_i and d_i are always multiplying each other in the initial condition

⁵Tanja Stadler and Sebastian Bonhoeffer. “Uncovering epidemiological dynamics in heterogeneous host populations using phylogenetic methods”. In: *Philosophical Transactions of the Royal Society B: Biological Sciences* 368.1614 (2013), p. 20120198.

MULTI-TYPE BIRTH DEATH MODELS⁶

Combining solutions on branches at ancestral nodes: let ancestral node A at time t connect to two branches, N and M
The solutions are combined in a more complicated expression than BiSSE:

$$D_{Ai}(t) = \sum_j (\lambda_{ij}(D_{Mi}(t)D_{Nj}(t) + D_{Mj}(t)D_{Ni}(t)))$$

⁶Tanja Stadler and Sebastian Bonhoeffer. "Uncovering epidemiological dynamics in heterogeneous host populations using phylogenetic methods". In: *Philosophical Transactions of the Royal Society B: Biological Sciences* 368.1614 (2013), p. 20120198.

MULTI-TYPE BIRTH DEATH MODELS⁷

At the root, at time t_0 , there is a collection of conditional likelihoods $D_{Ni}(t_0)$. Stadler and Bonhoeffer combine these in a similar manner as BiSSE, but they condition on observing one sampled individual,

$$\sum_i f_i \frac{D_{Ni}(t_0)}{1 - E_i(t_0)},$$

and they select f_i to be the expected fraction of lineages in state i under the demographic model

⁷Tanja Stadler and Sebastian Bonhoeffer. “Uncovering epidemiological dynamics in heterogeneous host populations using phylogenetic methods”. In: *Philosophical Transactions of the Royal Society B: Biological Sciences* 368.1614 (2013), p. 20120198.

GENERAL PHYLOGENETIC BIRTH-DEATH MODELS⁸

This model can be elaborated upon even further by including

- migration q_{ij}
- adding sampling rates that vary by state and/or time, $\psi_i(t)$
- including time-dependence in other rates, such as $\lambda_{ij}(t)$ (i.e. piece-wise constant functions)
- fossils (a special type of ancestral sampling event)
- mass extinction events (in macroevolution)
- various forms of conditioning (likelihood of a tree conditional on one lineage surviving to the present, etc.)

⁸Ailene MacPherson et al. “Unifying phylogenetic birth–death models in epidemiology and macroevolution”. In: *Systematic Biology* 71.1 (2022), pp. 172–189.

BDMM-PRIME

The relatively new BDMM-prime package in BEAST 2, accommodates all of these, and also samples demographic trajectories alongside phylogenies⁹

⁹Timothy G Vaughan and Tanja Stadler. “Bayesian Phylodynamic Inference of Multitype Population Trajectories Using Genomic Data”. In: *Molecular Biology and Evolution* 42.6 (2025), msaf130.

WHAT DO WE DO WITH A COLLECTION OF CHARACTER MAPPED TREES?

A BEAST2 analysis produces a posterior sample of character-mapped trees

With ancestral character estimation, we saw that we could compute a joint or marginal reconstruction of ancestral states on a fixed tree to understand likely trait changes in our data

With a collection of character-mapped trees, we identify a new challenge: how to summarize information about state changes across phylogenies that potentially have different topologies

This is an active area of research in our group. See <https://github.com/Lars-B/pyccd> for information about using conditional clade distributions to address this problem¹⁰

¹⁰Lars Berling and Caroline Colijn. “An Extended Clade Framework for Annotated Trees in the Context of Phylogeography and Transmission Tree Inference”. In: *bioRxiv* (2026), pp. 2026–04.

TABLE OF CONTENTS

- 1 MOVING TO A BAYESIAN FRAMEWORK
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MTBD IN R AND BEAST2

In R the following R package can simulate multi-type birth death models

- TreeSim

The following BEAST2 packages can simulate multi-type birth death models

- BDMM
- BDMM-prime